

A

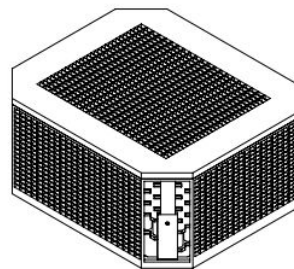
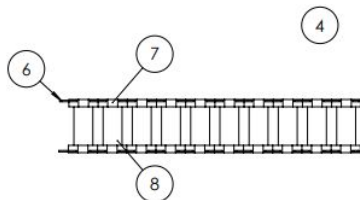
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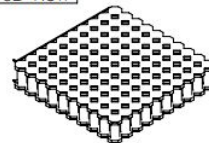
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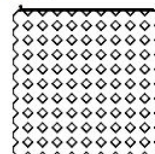
F



3D View



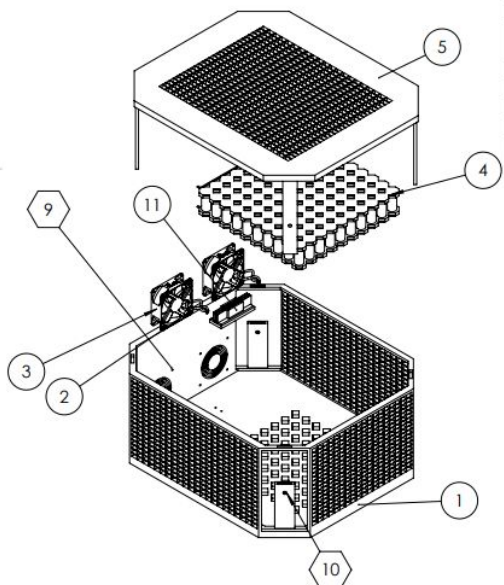
Side view



Top view



A3 FRAME
THIRD ANGLE PROJECTION



- RB system assembly instructions**
- 1) Install battery management system using 2xM4 screws
 - 2) Install fans using 4xM5 screws for each
 - 3) Install the wiring harness
 - 4) Slot in the battery holder using a push fit
 - 5) Slide lid prongs into opposite slots
 - 6) Screw 4xM5 screws into holes to securely attach top and bottom of casing

- Battery holder assembly instructions**
- 1) Slide bus bar in allocated slots
 - 2) Place all 100 batteries in the provided slots, resting them on top of the bus bar
 - 3) Slide second bus bar in allocated slots
 - 4) Place second assembled sheet on top of first, using push fit to ensure it's on securely

ITEM NO.	PART	PART NUMBER	QUANTITY
1	Main casing	ME1501/001/002/201	1
2	Wiring harness	ME1501/001/003/304	1
3	120mm fan	ME1501/001/003/301	2
4	Battery holder	ME1501/001/004	1
5	Main casing lid	ME1501/001/005/501	1
6	Battery holder bus bar	ME1501/001/004/402	2
7	Battery holder sheet	ME1501/001/004/401	2
8	Lithium ion 266550 battery	ME1501/001/004/403	100
9	M5x15 Screws	ME1501/001/002/203	8
10	M10x25 Screws	ME1501/001/002/202	4
11	Battery Management System	ME1501/001/003/305	1

Aston University
ASTON UNIVERSITY
ASTON TRIANGLE
BIRMINGHAM
B4 7ET
UNITED KINGDOM

TITLE: Casing Base

DRAWING NUMBER: ME1501/001/002/201

PROJECT: Battery casing

SHEET: 1 of 1

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SURFACE TREATMENT:
Ra 1.6µm

MATERIAL:
Aluminium 6061-T6

MACHINING DATA:
[]
ALL DIMENSIONS IN mm U.O.S

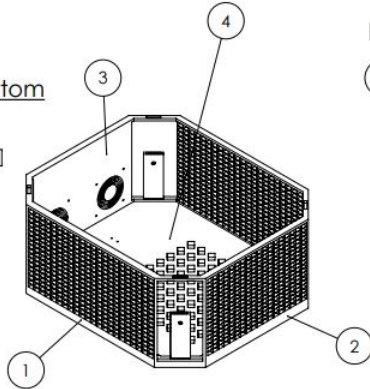
TOLERANCES:
NO DECIMAL PLACE 0.1
ONE DECIMAL PLACE 0.05
TWO DECIMAL PLACES 0.01
ANGLES 0.1

SCALE: 1:10
DRAWN: CHECKED: APPROVED:

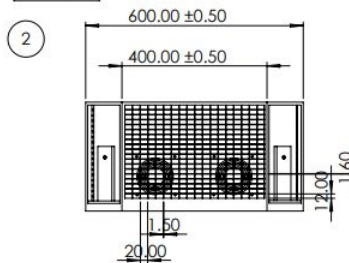


Casing bottom

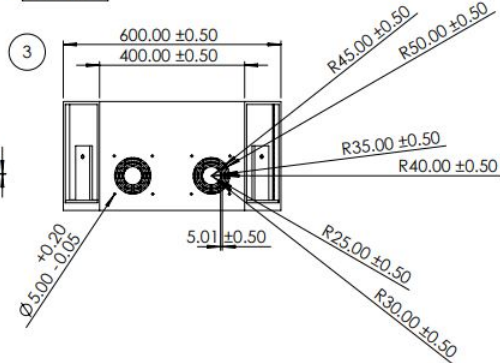
3D View



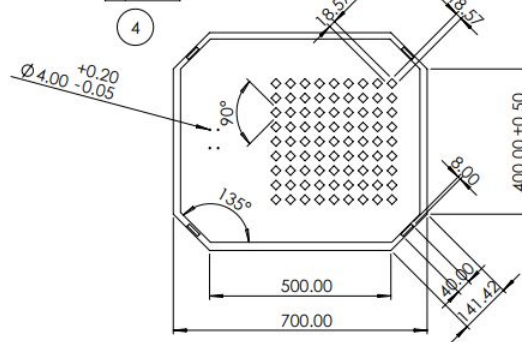
Front View



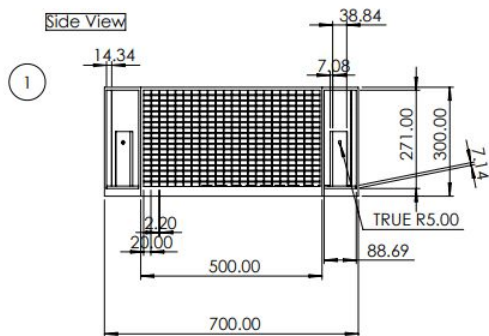
Back View



Top View



Side View



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SCALE
1:10

DRAWN
CHECKED
APPROVED

Aston University

ASTON UNIVERSITY
ASTON TRIANGLE
BIRMINGHAM
B4 7ET
UNITED KINGDOM

TITLE: Casing bottom

DRAWING NUMBER:
ME1501/001/002/201

PROJECT:
Battery casing

SHEET:
1 OF 1

General Assembly	Any Sub-Assembly collection of parts	Any Sub-Assembly collection of parts	UNIQUE PROJECT PART REF:	Item DESCRIPTION (insert as much info as possible to fully describe the item)	QTY	PART (or material) SUPPLIER		PART REF #		COST	COST	WEIGHT	WEIGHT		
						who will actually supply this part or material for manuf OR assembly labour	original manufacturer (if different)	SUPPLIERS PART REF #	ASTON REF # (if applicable EG. Drawing #)	Cost per item (individual)	Cost per item (total)	Weight per item (individual)	Weight per item (total)		
GEN ASSY	QTY	SUB-ASSY	QTY	SUB-ASSY	unique ref	PIECE PART DESCRIPTION									
ME2502/2025/X/GA /001	1	ME2502/2026/X/SA /001	1	ME2502/2026/X/SA /002	201	Outer casing base, CNC punched Aluminium	1	SSC Laser	/	/	/	£36	£36	0.80kg	0.80kg
					202	M10x25 screws and nuts for assembly	4	Screwfix	Easyfix	3569204689	/	£1.07	£4.28	0.025kg	0.10kg
					203	M5x15 screws and nuts for assembly	4	Screwfix	Worcester Bosch	8737603843	ME2502/2026/X/DTL/001	£1.05	£4.20	0.010kg	0.040kg
					Sub-Assembly process to SA drg		1	Assembly shop A	/	/	ME2502/2026/X/SA/002		£44.48		0.94kg
			1	ME2502/2026/X/SA /003	301	Axial cooling fan	2	RS Components	RS Pro	619-6804	/	£26.54	£53.08	0.15kg	0.30kg
					302	Fan mounting bracket	1	RS Components	RS Pro	221-7675	/	£5.30	£5.30	0.10kg	0.10kg
					303	Heat sink	1	RS Components	RS Pro	903-3087	ME2502/2026/X/DTL/003	£10.35	£10.35	0.20kg	0.20kg
					304	Wiring for fans and control unit	1	RS Components	Starstech	280-4793	/	£2.64	£2.64	0.040kg	0.040kg
					305	Battery control unit	1	RS Components	Tracopower	199-1775	/	£20.61	£20.61	0.25kg	0.25kg
					306	M4x20 screws for control unit	4	Screwfix	Worcester Bosch	5729205758	/	£1.02	£4.08	0.008kg	0.032kg
					Sub-Assembly process to SA drg		1	Assembly shop A	/	/	ME2502/2026/X/SA/003				
					Sub-Assembly process to SA drg		1	Assembly shop A	/	/	ME2502/2026/X/SA/001		£96.06		0.922kg
			1	ME2502/2026/X/SA /004	401	ALC4 V-battery holder sheet, injection moulded	2	Protolabs	/	/	/	£4	£8	0.29kg	0.58kg
					402	Nickel plated C11000 Copper bus bar	2	Powerbars Ltd	/	/	/	£10	£10	0.15kg	0.30kg
					403	Lithium ion 266550 battery (3.7V)	100	UltraXel	HRLenergy	HRL26650	/	£1.37	£80 in bulk	0.045kg	4.5kg
					Sub-Assembly process to SA drg		1	Assembly shop A	/	/	ME2502/2026/X/SA/004		£98		5.2kg
			1	ME2502/2026/X/SA /005	501	CNC punched aluminium body for outer casing part	1	SSC Laser	/	/	/	£42	£42	0.60kg	0.60kg
					202	M10x25 screws and nuts for assembly	2	Screwfix	Easyfix	3569204689	/	£1.07	£4.28	0.025kg	0.050kg
					Sub-Assembly process to SA drg		1	Assembly shop A	/	/	ME2502/2026/X/SA/005		£46.28		0.65kg
					Final Assembly process to GA drg		1	Assembly shop B	/	/	ME2502/2026/X/GA/001		£284.82		7.892kg